Retail Sales Analysis — Business Context and Problem Statement

1. Business Context

A retail company selling physical products through a dedicated sales team across three European markets.

- **Markets served:**
  - Germany
  - Czech Republic
  - Denmark
- **Sales channel:** A team of sales representatives operating across multiple towns within each country.
- **Period under review:** Fiscal years 2014 to 2017.
- **Recording level:** Performance captured continuously, at the level of individual transactions.

The core challenge:

The data describing this activity does not live in one system. It is scattered across several independent sources, each owned by a different part of the business.

- Sales records → exported as yearly files
- Product catalogue → operational database
- Geographic reference → published on the web
- Categories and sales roster → separate spreadsheets

In this fragmented state, the data offers little strategic value. The first requirement is to **consolidate these sources into a single, reliable model**, and from there answer the questions that matter.

2. Data Available

The analysis draws on **six core datasets**, each integrated from its own source.

Dataset	Description	Origin

Sales Transactions (FactSales)	Every sale — date, product, representative, location, units, discount, cost basis	Folder of yearly CSV files
Products	Product name, colour, retail price, standard cost, sub-category	PostgreSQL database
Geography	Country and town reference list	Published web table
Category	Top-level classification (Special, General)	Spreadsheet
Sub-Category	Links products to categories	Spreadsheet
Sales Representatives	Roster of sales staff	Spreadsheet

A dedicated **date dimension** is derived from the transaction history to support analysis across months, quarters, and years.

Together, the data answers:

- **What** was sold
- **At what** price and cost
- **Where** it was sold
- **By whom** it was sold
- **When** it was sold

3. The Business Problem

Despite holding rich transactional data, the business lacks a consolidated, trustworthy

view of its own performance.
Without that view, it cannot reliably identify:

- What is driving profit
- Which markets and products deserve investment
- How performance is trending over time

The analysis is therefore structured around **four strategic questions**:

3.1 Growth Velocity

- **Question:** Where and when did revenue see its most significant Month-over-Month and Quarter-over-Quarter inflection points?
- **Why it matters:**
  - Distinguishes genuine momentum from seasonal noise
  - Enables planning ahead of predictable peaks and troughs

3.2 Market Dominance and Volume

- **Question:** How does the leading market compare to smaller territories in revenue, units sold, and transaction efficiency?
- **Why it matters:**
  - Clarifies where the business is concentrated
  - Highlights where expansion opportunity lies

3.3 Portfolio Mix Analysis

- **Question:** Which product categories and price tiers (Low, Medium, High Value) deliver the strongest relative gross margin?
- **Why it matters:**
  - Identifies the most profitable segments of the catalogue
  - Directs marketing spend and inventory toward the greatest return

3.4 Human Capital Performance

- **Question:** Who are the core sales representatives anchoring bottom-line profitability?
- **Why it matters:**
  - Reveals where revenue is concentrated within the team

- Informs retention, recognition, and the sharing of effective practice

4. Objective

Build an **end-to-end analytical solution** that resolves the problem above:

1. **Consolidate** the diverse data sources into a single, well-structured model.
2. **Apply** a reliable calculation layer for revenue, cost, and profitability.
3. **Present** the findings through an interactive, navigable report.
4. **Deliver** clear, evidence-based answers to the four strategic questions, with actionable recommendations for the period ahead.